

New York TLC Project | ML Model

Executive Summary



Automatidata

Overview:

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To predict whether a customer will leave a generous tip on New York City TLC app, this summary shows results of building a random forest model based on several features.

Problem

The prediction results are very sensitive, since type I error might result in unhappy driver, while type II error might lead to customers dissatisfaction and can't be considered ethical.

Solution

The solution is to use f1-score as the basic metric since it accounts for both types of errors. Our model resulted in 74.5% f1-score, which is considered fair for predictions.

Details

- It was mandatory to do some features engineering to identify, extract and transform independent variables that are more likely to result in a successful model.
- At first, two models were developed, Random Forest and XGBoost classification models. After testing the models using different metrics, we found out that random forest model is more likely to perform better in this specific context (even though we lost our chance to see how our model will actually do with unseen data)
- The model showed 70.4% accuracy, 74.5% f1-score, 82.2% recall and 68% precision. Indicating relatively high false positive predictions.
- Most important features that affected customer tip were VendorID, mean distance, predicted fare, pickup location - especially location 138 - and mean duration. However the model does not clarify how these features affect the outcome and therefore more exploration and analysis of these parameters is needed.

Next Steps

- The model should be put into trial so that it can be more tuned and get clear feedback.
- It would be very useful if we collect more data about customer behavior with their final tip.